GSTBubble — Invoice & Calculator by Bubblesort

Complete Application Documentation
Version 1.0.0 · Flutter · Dart · Android/iOS/Web

1. App Overview

GSTBubble is a **GST Calculator & Invoice Generator** built for Indian small businesses. It runs entirely **offline** — all data stays on the user's device. No server, no cloud, no tracking.

What It Does

Feature	Description
GST Calculator	Instant tax breakdown (CGST/SGST/IGST) for any amount
Invoice Generator	Create professional PDF invoices with line items
Invoice History	Browse, search, re-download past invoices
Business Profile	Save your business details & logo for reuse

2. Architecture

[Architecture Diagram]
main.dart → MultiProvider → { ThemeProvider, GstCalculatorProvider, InvoiceProvider, BusinessProvider }
InvoiceProvider & BusinessProvider → DatabaseService (Hive)
HomeScreen → { Calculator Tab, Invoice Tab, History Tab, Business Tab }
Invoice Tab & History Tab → PdfService

Design Pattern: Provider + Service Layer

Layer	Role	Files
Screens	UI widgets, user interaction	`lib/screens/`
Providers	State management via ChangeNotifier	`lib/core/providers/`
Services	Database I/O, PDF generation	`lib/core/services/`
Models	Data classes (Invoice, Client, GstResult)	`lib/core/models/`
Utils	Currency formatting, legal dialog	`lib/core/utils/`
Theme	Light/Dark color system, typography	`lib/core/theme/`
Data	Static HSN code list	`lib/core/data/`

3. Project File Map

```
lib/
├── main.dart                          # App entry point, provider setup
├── core/
│   ├── data/
│   │   └── hsn_codes.dart             # 80+ common HSN codes
│   ├── models/
│   │   ├── client.dart                # Client + BusinessProfile models
│   │   ├── gst_result.dart            # GST calculation result model
│   │   └── invoice.dart               # Invoice + InvoiceItem models
│   ├── providers/
│   │   ├── business_provider.dart     # Business profile & client state
│   │   ├── gst_calculator_provider.dart # Calculator state
│   │   ├── invoice_provider.dart      # Invoice CRUD state
│   │   └── theme_provider.dart        # Light/Dark toggle
│   ├── services/
│   │   ├── database_service.dart      # Hive DB operations
│   │   └── pdf_service.dart           # PDF generation engine
│   ├── theme/
│   │   └── app_theme.dart             # AppColors, AppTheme (Light+Dark)
│   └── utils/
│       ├── currency_utils.dart        # Indian currency formatting
│       └── legal_dialog.dart          # Privacy Policy & T&C popup
└── screens/
    ├── home_screen.dart               # Bottom nav bar + tab switching
    ├── gst_calculator/
    │   └── gst_calculator_screen.dart # Calculator UI (~757 lines)
    ├── invoice/
    │   └── create_invoice_screen.dart # Invoice creation UI (~406 lines)
    ├── history/
    │   └── invoice_history_screen.dart # Invoice history + search
    └── profile/
        └── business_profile_screen.dart # Business setup UI
```

4. Packages & Dependencies

Package	Version	Purpose
`provider`	^6.1.2	State management across the app
`hive`	^2.2.3	Local NoSQL database (key-value)
`hive_flutter`	^1.1.0	Hive initialization for Flutter
`pdf`	^3.11.2	Programmatic PDF document building
`printing`	^5.13.4	PDF preview, share, print
`image_picker`	^1.1.2	Pick images from gallery/camera
`image_cropper`	^12.2.1	Crop, zoom, rotate images (UCrop)
`path_provider`	^2.1.5	Get app documents directory
`google_fonts`	^6.2.1	Inter & Poppins font families
`intl`	^0.20.2	Date formatting
`uuid`	^4.5.1	Unique ID generation
`flutter_launcher_icons`	^0.14.3	Custom app icon generation (dev)

5. App Workflow (Tab by Tab)

Tab 1: GST Calculator

[Calculator Flow]
Enter Amount → Select GST Rate (0/5/12/18/28% or custom) → Toggle Inclusive/Exclusive → Toggle Intra/Inter State → See live breakdown: Base, CGST, SGST, IGST, Total → "Add to Invoice" button → Creates invoice item with calculated values

How it works internally:

1. User types amount → `GstCalculatorProvider.setAmount()` triggers `_calculate()`
2. `GstResult.calculate()` factory does the math:
   - **Exclusive**: `baseAmount = amount`, `gst = amount × rate / 100`
   - **Inclusive**: `baseAmount = amount / (1 + rate/100)`, `gst = amount - base`
   - **Intra-state**: Splits GST into `CGST = gst/2` and `SGST = gst/2`

- **Inter-state**: Full amount goes to `IGST`

3. Provider calls `notifyListeners()` → UI rebuilds with new result
4. "Add to Invoice" calls `InvoiceProvider.addItemFromCalculator()` which auto-creates an invoice from BusinessProfile if none exists

Tab 2: Create Invoice

[Invoice Creation Flow]
"New" button creates invoice from BusinessProfile → From: Auto-filled from Business tab → To: Select saved client or type manually → Line Items: Name, HSN, Qty, Price, GST% → Invoice Summary: Subtotal, CGST/SGST/IGST, Total → "Generate Invoice" → Syncs latest BusinessProfile → Saves to Hive DB + increments counter → PdfService generates PDF → PDF preview / print / share

Key behaviors:

- **Sender card** reads live from `BusinessProvider.profile` via `context.watch`
- **Line items** support: custom Qty (including 0), custom Price, floating-point GST (e.g. 1.5%)
- **HSN helper** dialog lets you search 80+ common HSN codes
- **Inter-state toggle** per item controls CGST+SGST vs IGST split
- Before PDF generation, `_generateInvoice()` force-syncs all BusinessProfile fields into the invoice

Tab 3: Invoice History

[History Flow]
Monthly summary card (count, sales, tax) → Search bar (by invoice# or client) → Invoice cards list (swipe to delete) → Tap card → regenerate PDF | Status badges: Draft / Unpaid / Paid

How it works:

- `InvoiceProvider.loadInvoices()` reads all invoices from Hive, sorted by date descending
- `getMonthlySummary()` aggregates current month's totals
- `searchInvoices()` filters by invoice number or client name
- Re-downloading regenerates PDF from saved data via `PdfService`

Tab 4: Business Profile

[Business Profile Flow]
Logo Upload (Pick → Crop → Save locally) → Business Details (Name, Owner, GSTIN, PAN) → Contact Info (Phone, Email, Address) → Invoice Settings (Prefix, Default GST, Currency) → Payment Terms Template → Saved Clients List (Add/Edit/Delete) → Legal Section (Privacy Policy link)

Logo workflow:

1. `ImagePicker` opens gallery (max 1024×1024)
2. `ImageCropper` (UCrop) opens with crop/zoom/rotate — supports Square, 3:2, 4:3, 16:9, free
3. Cropped image is copied to `getApplicationDocumentsDirectory()` for permanent storage
4. Path saved to `BusinessProfile.logoPath` via Hive

Data persistence:

- Every field change calls `BusinessProvider.updateProfileField()` → instantly saves to Hive
- Data survives app restarts, cache clears, and updates

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6. Data Models

InvoiceItem

Field	Type	Default	Description
`id`	String	UUID	Unique identifier
`itemName`	String	''	Product/service name
`hsnCode`	String	''	HSN/SAC code
`quantity`	double	1	Quantity (supports decimals)
`unitPrice`	double	0	Price per unit
`gstRate`	double	18	GST percentage (supports 1.5%, etc.)
`isInterState`	bool	false	IGST if true, else CGST+SGST

Computed properties: `amount`, `gstAmount`, `cgst`, `sgst`, `igst`, `totalAmount`

Invoice

Field	Type	Description
`id`	String (UUID)	Unique invoice ID
`invoiceNumber`	String	Formatted: PREFIX-0001
`invoiceDate` / `dueDate`	DateTime	Dates
`senderName/Gstin/Address/Phone/Email`	String	From business profile
`clientName/Gstin/Address`	String	Bill-to details
`items`	List<InvoiceItem>	Line items
`status`	InvoiceStatus	draft / unpaid / paid
`paymentTerms` / `notes`	String	Footer text
`logoPath`	String?	Path to business logo

Computed: `subtotal`, `totalCgst`, `totalSgst`, `totalIgst`, `totalGst`, `grandTotal`, `gstBreakdown`

BusinessProfile

Stores: businessName, ownerName, gstin, pan, phone, email, address, logoPath, invoicePrefix, nextInvoiceNumber, defaultGstRate, currency, paymentTerms

Client

Stores: id, name, gstin, address, phone, email

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7. Database (Hive)

Hive is a lightweight, pure-Dart NoSQL database. Data is stored in **Boxes** (like tables).

Box Name	Key	Value	Purpose
`settings`	'businessProfile'	Map	Business profile data
`invoices`	Invoice UUID	Map	Invoice header data
`invoice_items`	Invoice UUID	List<Map>	Line items per invoice
`clients`	Client UUID	Map	Saved client records

Invoice numbering: The `nextInvoiceNumber` field lives inside `BusinessProfile`. When an invoice is saved for the first time, `DatabaseService.incrementInvoiceNumber()` bumps the counter by 1 and persists it back to the profile.

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8. PDF Generation

`pdf_service.dart` generates a professional A4 invoice PDF.

PDF Layout Structure

```
┌──────────────────────────────────────────┐
│  [Logo] Business Name        TAX INVOICE │  ← Dark header
│         GSTIN, Address       INV-0001    │
├────────────────────┬─────────────────────┤
│  Invoice Details   │  Bill To            │  ← Light grey boxes
│  No, Date, Due     │  Client, GSTIN, Addr│
├────────────────────┴─────────────────────┤
│  # | Item | HSN | Qty | Price |GST%| ... |  ← Items table
│  1 | ...  | ... | ... |  ...  |C+S | ... |
│  2 | ...  | ... | ... |  ...  |IGST| ... |
├──────────────────────────────────────────┤
│  Amount in Words    │  Subtotal          │  ← Totals
│  Rupees ... Only    │  CGST @ 9%  Rs.xxx │
│                     │  SGST @ 9%  Rs.xxx │
│                     │  IGST @ 18% Rs.xxx │
│                     │  ┌──────────────┐  │
│                     │  │ Grand Total  │  │
│                     │  └──────────────┘  │
├──────────────────────────────────────────┤
│  Payment Terms & Notes                   │
├──────────────────────────────────────────┤
│  Thank you for your business!            │  ← Footer
│  Phone: ... | Email: ...                 │
└──────────────────────────────────────────┘
```

Key PDF Features

- **Logo embedding**: Reads from local file path or asset, renders as 48×48 rounded image
- **GST Type in table**: Shows "C+S" (CGST+SGST) or "IGST" label per item
- **Detailed tax breakdown**: Summary splits into CGST @ X%, SGST @ X%, IGST @ Y% rows
- **Amount in words**: `numberToWords()` converts to Indian English ("Rupees ... Only")
- **Currency**: Uses "Rs." prefix (avoids ₹ glyph issues in PDF fonts)
- **Fonts**: Inter (body) + Poppins (headings) loaded via `PdfGoogleFonts`

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9. Theme System

`app_theme.dart` defines the complete design system.

Token	Light	Dark
Background	`#F5F7FA`	`#0D1117`
Card	`#FFFFFF`	`#1C2333`
Primary Text	`#0A1628`	`#F0F6FC`
Accent	`#FF6B35` (Coral)	`#FF6B35` (Coral)
CGST color	`#6366F1` (Indigo)	same
SGST color	`#8B5CF6` (Purple)	same
IGST color	`#EC4899` (Pink)	same

Fonts: Poppins (headings), Inter (body) — both via Google Fonts.

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10. Currency Utilities

`currency_utils.dart`

Function	Input	Output
`formatIndianCurrency(123456.50)`	double	"Rs. 1,23,456.50"
`numberToWords(12500.0)`	double	"Rupees Twelve Thousand Five Hundred Only"

Uses Indian numbering: last 3 digits, then groups of 2 (e.g., 1,23,45,678).

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11. Legal Compliance

- **Info button (ℹ️)** present in AppBar of all 4 tabs
- Shows `showAppLegalDialog()` — a scrollable AlertDialog with:
  - **Privacy Policy**: No data collection, all local storage
  - **Terms & Conditions**: Disclaimer for calculation accuracy
- Standalone files: `PRIVACY_POLICY.md`, `TERMS_AND_CONDITIONS.md` in project root

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12. Android Configuration

AndroidManifest.xml

```
<activity android:name=".MainActivity" ... />
<activity
    android:name="com.yalantis.ucrop.UCropActivity"
    android:screenOrientation="portrait"
    android:theme="@style/Theme.AppCompat.Light.NoActionBar"/>
```

The `UCropActivity` is **required** by `image_cropper` — without it, the app crashes when opening the crop screen.

App Icon

Configured via `flutter_launcher_icons` in `pubspec.yaml`, using `assets/images/GSTBubbleLogo.png`.

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13. Build Commands

Command	Output
`flutter run`	Debug on connected device
`flutter run -d chrome`	Debug on Chrome (web)
`flutter build apk`	Release APK (~68 MB)
`flutter build apk --split-per-abi`	Smaller per-architecture APKs
`flutter build appbundle`	AAB for Play Store
`flutter install`	Install APK on connected device

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14. Data Flow Summary

[Data Flow Sequence]

User → Screen: Fill business details
Screen → Provider: updateProfileField()
Provider → Hive DB: saveBusinessProfile()

User → Screen: Create new invoice
Screen → Provider: createNewInvoice()
Provider → Hive DB: getBusinessProfile()
Provider → Screen: Invoice with sender details

User → Screen: Add line items
Screen → Provider: addItemToInvoice()

User → Screen: Tap "Generate Invoice"
Screen → Provider: Sync latest BusinessProfile
Provider → Hive DB: saveInvoice() + incrementInvoiceNumber()
Screen → PdfService: generateInvoicePdf(invoice)
PdfService → Screen: PDF bytes
Screen → User: Preview / Print / Share